

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

VI Semester of BSc (Biological Sciences) Examination April-May 2018

Course code & Name: MI 601 Fermentation Technology

Date: 30/04/2018 Day: Monday Time: 1:30 P M To 1:50 P M Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions. 10

1. The production curve of primary metabolite is similar to that of the following of microbes.
(i) log phase (ii) death phase (iii) growth curve (iv) stationary phase
2. For citric acid production, fermentation medium should contain the following ions in limiting concentration
(i) Fe^{+2} and Mg^{+2} (ii) Na^{+} and K^{+} (iii) riboflavin (iv) biotin
3. To make ethanol production from lignocellulose economically viable, the producing organism should be able to ferment the following sugar to ethanol
(i) glucose (ii) sucrose (iii) xylose (iv) xylan
4. The following mutants of *Streptomyces* sp over produce streptomycin
(i) Factor A⁺ (ii) Factor B⁺ (iii) Factor A⁻ (iv) Factor B⁻
5. The following is the inducer of alkaloid production by *Claviceps purpurea*
(i) Tryptophan (ii) Elymoclavine (iii) Riboflavin (iv) Lysergic acid
6. The following compound is added continuously for the production of benzyle penicillin, due to its toxicity at high concentration
(i) benzene (ii) phenyl acetic acid (iii) acetic acid (iv) benzoic acid

- Page 2 of 3

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Course code & Name: MI 601 Fermentation Technology

Date: 30/04/2018 Day: Monday Time: 1:50 P M To 3:30 P M Maximum Marks: 25

Instructions:

1. Section I and II must be attempted in SINGLE ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION - I

- Q - II Answer the following questions as directed 10**
- A Differentiate between any one the following (02)**
- i. Primary and secondary metabolites giving examples
 - ii. Fermentation and Biotransformation
- B Discuss briefly any two of the following (06)**
- i. Fermentation for penicillin production
 - ii. Therapeutic Uses of alkaloids
 - iii. Characteristics of microorganisms used for SCP
- C Justify Though Mycobacterium, accumulate up to 30-40% of lipids, it is not used as SCO (02)**

SECTION - II

- Q-III Answer the following questions as directed 15**
- A Discuss briefly any two the following (04)**
- i. Factors affecting ethanol recovery during ethanol fermentation
 - ii. Strain improvement strategies for amino acid production with example
 - iii. Downstream processing of lactic acid
- B Discuss briefly any one the following (03)**
- i. Microbial consortium involved in biogas production
 - ii. Microbiological methods for the production of amino acid with examples
- C Attempt any three of the following (06)**
- i. Give an over- view of biofertilizer production technology
 - ii. Discuss briefly the fermentation for enzyme production
 - iii. Discuss briefly: Mode of Action of Bt Toxin
 - iv. Discuss briefly: Production of semisynthetic penicillin
- D Justify Integrated product recovery system during butanol fermentation leads to high yields (02)**